Night shift — full-development SHD comparison

ar068 · 18 July 2026 · Draft

Digest

`exp068` completed the locked one-seed comparison without a protocol change. PING reached 45.72% official-test accuracy against COBA's 35.07%, a registered difference of +10.64 percentage points and therefore a promising exploratory PING accuracy claim. PING also had lower official-test cross-entropy (2.06 versus 2.35), higher macro-class accuracy (45.47% versus 34.64%), and lower excitatory firing (10.50 Hz versus 28.44 Hz). Both checkpoints froze before the complete 2,264-utterance official test was exposed. This is a one-seed result that motivates, but does not substitute for, confirmation.

Mandate

Purpose and scope

This one-seed exploratory shift asks whether PING's modest point advantage in `exp066` becomes clearer when both architectures train on substantially more Spiking Heidelberg Digits (SHD) data under a leakage-resistant train/validation/test protocol. It is intended to identify a claim worth defending later, not to provide multi-seed confirmatory evidence.

The primary question is whether PING outperforms matched COBA in overall official-test accuracy. Accuracy relative to excitatory activity is reported transparently as paired accuracy and firing-rate measurements; no post-hoc composite score will be invented.

Registered experiment

Field	exp068
Hypothesis	With matched training and readout settings on substantially more SHD development data, PING exceeds COBA by at least two percentage points in overall official-test accuracy while using less excitatory spike activity.
Baseline	exp066 established one-seed feasibility on 1,000 training utterances: COBA reached 31.0% best held-out accuracy and PING 34.7%. The present contemporaneous matched COBA cell is the architectural baseline; the 5% SHD chance floor remains a diagnostic reference.
Kill criterion	The accuracy hypothesis is falsified if PING does not exceed COBA on overall official-test accuracy. Kill either cell for non-finite loss or logits, persistent skipped updates, silence, clear saturation, or failure to complete the fixed protocol; kill the comparison for different partitions, mismatched

Field	exp068
	registered settings, or any use of official-test data before both checkpoints are frozen.
Seed count	1
Budget	One local plumbing smoke plus one 40-epoch full run per architecture; two concurrent pods maximum, 3 h target per pod, and 10 USD hard total RunPod spend.
Status	done

Data partition and sealed evaluation

Use SHD's official 8,156-utterance training split and 2,264-utterance test split. At seed 42, partition the official training split deterministically into approximately 90% development training and 10% validation, stratified jointly by class and speaker. COBA and PING receive byte-identical index sets, preprocessing, and sample-ordering policy.

The complete official test split remains sealed during implementation, debugging, training, hyperparameter selection, stopping, and checkpoint selection. After both training runs have finished and both checkpoint choices are frozen, evaluate each selected checkpoint exactly once on the full official test set. Neither result may be revealed or acted upon before both checkpoints are frozen. Overall official-test accuracy is primary. Macro class accuracy and accuracy for seen versus unseen speakers are registered diagnostics; SHD test speakers 4 and 5 form the unseen-speaker group.

Matched architectures and optimisation

Both cells use 256 excitatory and 64 inhibitory cells, Dale-constrained fixed recurrent weights, input-weight mean 0.9, 95% input sparsity, membrane-mean readout with scale 225, Adam learning rate 0.0004, batch size 32, 1 ms integration timestep, 1,000 ms simulation duration, and no firing-rate regulariser. Initialisation, data, preprocessing, ordering, input weights, readout, and every other applicable setting remain matched.

COBA disables the inhibitory loop and uses no voltage-gradient dampening. PING enables the fixed inhibitory loop and uses voltage-gradient dampening
1000. No architecture-specific tuning or rescue is permitted after either

result is seen.

Training and checkpoint selection

First run a small local staged train/validation smoke test to verify the partition, data plumbing, tensor shapes, finite optimisation, checkpointing, and active populations. Smoke measurements are not scientific evidence.

Each full cell then trains for exactly 40 epochs. Select the checkpoint with highest validation accuracy; break equal-accuracy ties with lower validation cross-entropy and then the earlier epoch. The experiment runner must preserve this rule explicitly rather than relying on an incompatible default. Only after both checkpoint identities are recorded may the sealed official-test evaluation begin.

Outcomes and interpretation

The primary exploratory result is $\delta_{\text{accuracy}} = \text{PING accuracy} - \text{COBA accuracy}$ on the complete official test split.

1. If the difference is at least +2 percentage points, the result identifies a promising PING accuracy claim for a later confirmatory study.
2. A positive difference below +2 points is a weak directional signal, not a useful accuracy claim.
3. A zero or negative difference provides no PING accuracy advantage under this recipe.

Registered secondary diagnostics are official-test cross-entropy, validation accuracy and loss curves, macro class accuracy, seen- and unseen-speaker accuracy, excitatory and inhibitory firing rates, accuracy alongside excitatory spike rate, runtime, peak GPU memory, non-finite batches, skipped updates, and directly matched input/E/I rasters for preselected official-test utterances.

Integrity and stopping rules

Kill and record a cell if loss or logits become non-finite, skipped updates persist, activity is silent or clearly saturated, or the fixed protocol cannot complete. Kill the comparison if the cells receive different partitions or any registered setting differs. Failed and killed attempts are retained rather than overwritten. The question, baseline, primary outcome, interpretation bands, and kill criteria cannot change after results are observed.

Operating contract

```
budgets:
  wall_clock_target_per_pod: 3h
  wall_clock_per_shift: 8h
  runpod_total_usd: 10
  max_pods_concurrent: 2
  seeds_default: 1
scope:
  collection: spiking-heidelberg-digits
  may_propose: true
  may_edit_snn_cli: false
  experiment_id: exp068
  activity_trace: ar069
stop_when:
  - queue_empty
  - shift_budget_exhausted
  - runpod_budget_exhausted
  - protocol_integrity_failure
  - build_red_twice
```

Two concurrent RunPod pods are authorized: one for COBA and one for PING, preferably using RTX 5090 or an equivalently suitable GPU. Total experiment spend may not exceed 10 USD. Both pods must be reaped immediately after collection or failure. Stop for additional authority before exceeding the budget or pod limit, editing `tools/snn`, changing this design, or taking any other action outside the approved scope.

Implementation should use `experiments/exp068.py` and existing CLI capabilities. Runner-controlled staging of deterministic SHD train/validation files is permitted when provenance is retained and the untouched official test split is used only for final evaluation. Use focused lightweight checks and `demolab build`; do not run the full test suite on the 4 GB host.

Record

2026-07-18 22:50–22:52 UTC: mandate lock and night branch
The scientist approved the complete mandate without amendment. It was preregistered on `main` at commit `2ee03ac` before the experiment runner or any exp068 result existed. The dedicated night branch was created from that exact commit. At the compute gate, exp068 had spent 0 USD and no pods were active.

2026-07-18 22:52–23:04 UTC: implementation and local smoke
The experiment runner stages the official training data into a deterministic 7,340-utterance development-training set and an 816-utterance validation set, stratified jointly by speaker and class at seed 42. During training, the official test split is physically unavailable. Experiment-side checkpoint instrumentation preserves the registered validation tie-break without changing `tools/snn`.

The registered local smoke used 128 staged training and 128 staged validation utterances for two epochs. Both cells remained finite and active, and neither recorded a skipped update or non-finite batch. COBA selected epoch 1 with 11.72% validation accuracy and 23.93 Hz excitatory activity. PING selected epoch 2 with 10.16% validation accuracy, 2.95 Hz excitatory activity, and 14.56 Hz inhibitory activity. These values test plumbing only and are not evidence for the hypothesis. The host's complete 8,156-sample training and 2,264-sample official-test caches were restored byte-for-byte after staging.

2026-07-18 23:11 UTC: full comparison dispatched
A final evaluator audit used two staged validation utterances and the smoke checkpoint to verify the 20-class checkpoint load, prediction capture, cross-entropy, activity, speaker grouping, and runtime paths without accessing official-test evidence. Two RTX 5090 pods then started concurrently from pinned commit `96748e5`, one per architecture, at 0.99 USD per pod-hour. Each has a 10,800-second self-removal backstop, for a maximum actual-rate exposure of 5.94 USD. The pre-dispatch spend and active-pod count were both zero. The official-test seal remains closed until both validation checkpoints freeze.

2026-07-19 00:24–00:26 UTC: paired freeze, sealed test, and reap
PING froze epoch 40 at 43.50% validation accuracy; COBA froze epoch 25 at 40.81%. Both frozen records carried the same development-training and validation index hashes. Only after both records existed did the cells stage and evaluate the complete official test. The paired result was then disclosed: PING classified 1,035 of 2,264 utterances (45.72%) and COBA classified 794 (35.07%), for the registered +10.64 percentage-point difference. PING's macro class accuracy was 45.47% versus 34.64%; seen-speaker accuracy was 44.10% versus 38.68%; unseen-speaker accuracy was 46.09% versus 34.24%.

Both runs completed with zero skipped updates and zero non-finite forward batches. PING's official-test E/I rates were 10.50/47.99 Hz; COBA's E rate was 28.44 Hz and its disabled inhibitory loop remained at zero. The pods self-removed immediately after their completion markers, an explicit reap found nothing left to terminate, and a provider listing confirmed zero active pods. Observed compute spend was 2.4414148 USD: 1.220103775 USD for COBA and 1.221311025 USD for PING at the actual 0.99 USD/h rate.